HEIMDALL

Host Environment Intelligence for Monitoring, Diagnostics and Anomaly Learning on Linux

24CYS305 – Artificial Intelligence and Neural Networks

Team Details

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Project Idea and Problem Statement

Linux systems continuously generate telemetry such as CPU utilization, memory usage, disk activity, network activity, process information, system load, services, and security related events. Existing Linux monitoring tools provide access to this information, but interpreting multiple metrics and identifying the actual cause of abnormal behavior often requires considerable Linux and system administration knowledge.

HEIMDALL proposes an AI powered system intelligence framework that analyzes Linux system telemetry to understand the current operating state, identify performance or security related issues, detect abnormal behavior, and provide intelligent troubleshooting and decision support.

Instead of functioning only as a monitoring dashboard, HEIMDALL aims to transform raw Linux telemetry into meaningful system intelligence by answering three major questions:

- What is currently happening in the system?
- Is the observed behavior normal or abnormal?
- What should the user investigate next?

Proposed Approach and AI Concepts

HEIMDALL will collect selected Linux telemetry including CPU, memory, disk, network, process, and system event information. The collected data will be cleaned, normalized, and converted into features suitable for neural network analysis.

A **Multi Layer Perceptron (MLP)** will be developed to classify known operating conditions such as normal operation, CPU stress, memory pressure, disk pressure, and selected system abnormalities. The model will demonstrate feed forward propagation, backpropagation, ReLU and Softmax activation functions, cross entropy loss, the Adam optimizer, and regularization techniques such as Dropout and Early Stopping.

As an advanced feature, an Autoencoder based anomaly detection model will be explored. The Autoencoder will learn patterns associated with normal Linux behavior and use reconstruction error to identify telemetry that significantly deviates from the learned baseline.

The proposed workflow is:

Linux Telemetry → Preprocessing → MLP / Autoencoder → System Analysis → Diagnosis → Decision Support

Uniqueness

The key difference between HEIMDALL and conventional Linux monitoring tools is that the proposed system focuses on interpreting the system behavior rather than simply displaying metrics.

For example, instead of only reporting high CPU usage, HEIMDALL can analyze CPU usage together with system load, memory, processes, and other telemetry to determine whether the behavior represents a known problem or an unusual operating condition.

The project uniquely combines **Linux telemetry monitoring, neural network based system state classification, anomaly detection, performance/security analysis, and intelligent troubleshooting** within a single framework. HEIMDALL is designed primarily as a decision support system, keeping the user in control instead of providing unrestricted autonomous control over Linux.

Literature Review

Recent research in Linux system monitoring and security shows an increasing use of eBPF, kernel level telemetry, neural networks, and machine learning based anomaly detection. These developments are closely related to the proposed HEIMDALL framework.

The work *"Real Time Intrusion Detection and Prevention with Neural Network in Kernel Using eBPF"* investigates the execution of neural network based intrusion detection within the Linux kernel using eBPF. The approach aims to reduce the latency and context switching overhead associated with transferring telemetry to user space machine learning systems. This work is relevant to HEIMDALL because it demonstrates the possibility of combining low level Linux telemetry with neural network based security analysis.

The study *"An In-Depth Analysis of eBPF Based System Security Tools in Cloud-Native Environments"* analyzes eBPF based monitoring and security frameworks such as Falco, Tetragon, KubeArmor, and Tracee. It provides insight into the architecture and performance of modern Linux telemetry and runtime monitoring systems. This is relevant to HEIMDALL's telemetry collection and system monitoring architecture.

Retriever: A Distributed Intrusion Detection System for NOS Enabled Networks explores lightweight anomaly detection using low level host events and provenance information. The system generates behavioral representations and calculates anomaly scores to identify suspicious activities. This supports HEIMDALL's objective of learning system behavior and detecting abnormal operating conditions.

Research on *eBPF-Based Runtime Detection of Semantic DDoS Attacks in Linux Containers* demonstrates the use of eBPF telemetry, system call frequencies, and network information for identifying abnormal behavior in Linux container environments. Similarly, *KubeRosy* investigates dynamic system call filtering using eBPF for container runtime protection.

These studies demonstrate the growing importance of kernel level telemetry, eBPF, behavioral analysis, and machine learning for Linux monitoring and security. HEIMDALL builds on these concepts by combining Linux system telemetry with neural network based system state classification and anomaly detection to provide both performance and security intelligence together with intelligent troubleshooting and decision support.

Expected Outcome

The expected outcome is a working Linux prototype capable of collecting system telemetry, identifying known operating states, detecting potentially abnormal behavior, highlighting relevant performance or security concerns, and providing meaningful troubleshooting information to users.

The project will demonstrate a practical integration of **Artificial Intelligence, Neural Networks, Operating Systems, and Cybersecurity**.

References

1. Real Time Intrusion Detection and Prevention with Neural Network in Kernel Using eBPF," *IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2024–2025.
2. "An In Depth Analysis of eBPF Based System Security Tools in Cloud-Native Environments," *IEEE Access*, vol. 13, 2025.
3. "Retriever: A Distributed Intrusion Detection System for NOS Enabled Networks," *IEEE Transactions on Dependable and Secure Computing*, 2026.
4. "eBPF-Based Runtime Detection of Semantic DDoS Attacks in Linux Containers," *IEEE Access / IEEE Conference on Data and Cloud*, 2025.
5. "KubeRosy: A Dynamic System Call Filtering Framework for Containers," *IEEE Access*, 2024–2025.

Table 1: Research Papers Relevant to the Proposed Work

No.	Paper Title	Authors	Year	Relevance to Research Topic
1	Aquila: Efficient In Kernel System Call Telemetry for Cloud Native Environments	Jaehyuk Shin	2025	Demonstrates how in kernel eBPF ring buffers intercept, filter, and correlate low-level system call streams dynamically with minimal CPU overhead, solving telemetry ingestion bottlenecks for high frequency Linux workloads.
2	A Flow Based IDS Using Machine Learning in eBPF	Maximilian Bachl, Joachim Fabini, and Tanja Zseby	2021	Explores embedding lightweight machine learning decision trees directly inside eBPF kernel bytecode for early stage security filtering before telemetry reaches user space engines.
3	DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning	Min Du, Feifei Li, Guineng Zheng, and Vivek Srikumar	2017	Foundational work modeling Linux system logs and system call sequences using LSTMs to detect behavioral anomalies.
4	U-SCAD: An Unsupervised Method of System Call Driven Anomaly Detection for Containerized Edge Clouds	Jiawei Ye et al.	2025	Uses Syscall2vec embeddings to represent system call parameters as vectors for unsupervised anomaly detection in containerized Linux environments.
5	Provenance Based Intrusion Detection: Opportunities and Challenges	Xueyuan Han, Thomas Pasquier, and Margo Seltzer	2018	Introduces provenance graphs for modeling process, file, and socket interactions to trace attacks and privilege escalation.
6	Anomaly Detection Using Graph Neural Network on System Provenance Data	Bhupendra Lakha	2024	Applies Graph Neural Networks to Linux provenance graphs for detecting structural anomalies and zero day attacks.
7	Deep Learning for Anomaly Detection: A Survey	Raghavendra Chalapathy and Sanjay Chawla	2019	Surveys Autoencoders, VAEs, GANs, and other deep learning approaches for anomaly detection in high dimensional telemetry.
8	LogEDL: Log Anomaly Detection via Evidential Deep Learning	Yaokun Duan, Kaiwen Xue, Hao Sun, et al.	2024	Introduces Evidential Deep Learning to estimate prediction confidence, reducing false positives in log anomaly detection.
9	Multi Modal Anomaly Detection in Sensor Derived Metadata Using Instruction Tuned Transformers	Khalid M. Alhawiti	2025	Demonstrates multimodal fusion of system logs and numerical metrics for enhanced anomaly detection.
10	AI Powered Data Pipeline Observability and Root Cause Analysis	ResearchGate Community / Industry Observability Review	2026	Combines AI anomaly detection with dependency graphs to automate root cause analysis and explain system alerts.